Step 0: Initial Document Check

• **0.1:** The document is an RCC structural drawing of "FOUNDATIONS" only.
• **0.2:** The site location is "NH OFFICE".
• **0.3:** All compliance checks will be based only on IS 456:2000 and SP 34.

Step 1: Locate the "NOTES" Section

• **Status:** Compliant

Step 2 & 3: Extract and Verify Design Parameters from "NOTES"

Criteria	Extracted Value	Compliance Check	Status
1. Grade of Concrete	M25 for all R.C.C works.	M25 is a suitable grade for general RCC works. Without specific environmental exposure conditions for "NH OFFICE", it's difficult to definitively confirm full compliance with IS 456:2000 Table 3 and 5. However, M25 is a common and generally acceptable grade.	Cannot Verify
2. Reinforcement Bars	HYSD TMT BARS of Grade Fe 500 conforming to IS 1786-1985. Shear reinforcement and stirrups are indicated by Y8@150 C/C and Y8@200 C/C in the lift pit section and Y8@200 for column ties.	Fe 500 is a standard and compliant grade. The mention of Y8@150 C/C and Y8@200 C/C for shear reinforcement/stirrups is noted.	Compliant
3. Lap Length	50 times dia of the bar and to be staggered such that not more than 50% of the bars are lapped at a section.	Compliant with IS 456:2000 and SP 34.	Compliant
4. Clear Cover	Footing/Wall: 50 mm, Columns: 40 mm, Slab: 20 mm, Beam: 25 mm.	Footing cover of 50mm is the minimum as per IS 456:2000. If 70-75mm is not used, PCC is required, which is specified as "5" THK. PCC" in Section X1. Column cover of 40mm is compliant. Slab and Beam covers are also compliant.	Compliant
5. Development Length (Ld)	50 times the dia. of the bar.	Compliant with IS 456:2000 and SP 34.	Compliant
6. Safe Bearing Capacity (SBC) of soil	12.5 T/m2. PCC is mentioned as "5" THK. PCC" in Section X1.	SBC is mentioned. PCC is specified.	Compliant
7. Seismic Zone and Wind Load	Structure is designed for Seismic Zone III and Wind Forces as per IS 1893:2016 & IS 875:2015 respectively.	This information is mandatory and provided.	Compliant
8. Building Limitations	Structure is designed for Ground +2 Stories only.	Building limitations are mentioned.	Compliant

9. Structure's Purpose	NH OFFICE (Foundation Layout).	The purpose is mentioned as "NH OFFICE" and the drawing title is "FOUNDATION LAYOUT".	Compliant
10. Floor Heights	First Floor: 3.6M, Second Floor: 3.6M, Terrace: 3.6M (from plinth beam).	Floor heights are mentioned.	Compliant
11. Schedule of Footings	"SCHEDULE OF FOOTINGS" table is present and consistent with the drawing.	The table is present and appears consistent.	Compliant
12. Footing Type	Isolated footings (F1) and Raft footings (RF1, RF2) are shown.	Isolated footings are used for a G+2 storey building, which is a low-to-medium rise building. Raft footings are also present.	Compliant
14. Raft Foundation Reinforcement	RF1: Top Y12@125, Bottom Y16@150. RF2: Top Y12@125, Bottom Y16@125.	Both top and bottom reinforcement are specified for raft foundations.	Compliant
15. Lift Design	A lift pit is shown with a depth of "1500 MIN. OR AS PER ARCH." and reinforcement details. Bending of steel reinforcement at the footing base is shown in Section X.	Lift pit depth is specified as minimum 1500mm. Bending details are shown.	Compliant
16. Soil Improvement	"90 TO 40MM SIZE BOULDERS COMPACTED WITH SAND/QUARRY DUST" is mentioned below the footing in Section X.	Soil improvement details are provided.	Compliant
17. Column Ties	Column ties are shown in the column schedules (e.g., Y8@200). They appear continuous in the typical column sections.	Column ties are specified and appear continuous.	Compliant
18. Plan of Ties	A separate plan for ties is not explicitly present, but tie details are provided within the column schedules.	While a separate plan is not present, the details are provided in the column schedules.	Cannot Verify
19. Outer Ties Check	Column schedules specify ties (e.g., Y8@200). Percentage of steel in columns is not explicitly stated in the notes but can be calculated from the column schedules. Footing/slab reinforcement is specified.	The percentage of steel in columns (>= 0.8%) and footings/slabs (>= 0.12%) is not explicitly stated in the general notes. This would require calculation from the provided bar sizes and column/footing dimensions.	Missing Information
20. Cross-Section Area	Not explicitly mentioned whether gross cross-section area is used for columns/footings and effective area for slabs.	This information is not specified in the notes.	Missing Information
21. Steel Curtailment	The building is G+2 stories. Steel curtailment is not explicitly mentioned in the notes.	For a G+2 building, curtailment might not be as critical as for high-rise, but it's good practice to mention. It's not explicitly detailed.	Missing Information

22. Maximum Steel Percentage in Columns	Not explicitly mentioned in the notes.	This value is not stated in the notes.	Missing Information

Step 5: Report Missing or Wrong Information

Grade of Concrete: Cannot be fully verified without specific environmental exposure conditions for "NH OFFICE".

Outer Ties Check: Percentage of steel in columns and footings/slabs is not explicitly stated in the general notes.

Cross-Section Area: Not explicitly mentioned whether gross cross-section area is used for columns/footings and effective area for slabs.

Steel Curtailment: Not explicitly detailed for the G+2 storey building.

Maximum Steel Percentage in Columns: Not explicitly mentioned in the notes.

Summary of Compliance

• **Total Criteria Evaluated:** 22

• **Compliant Items:** 13

• **Non-Compliant Items:** 0

• **Missing Information Items:** 4

• **Cannot Verify Items:** 2

• **Not Applicable Items:** 1

• **Overall Verdict:** The drawing is largely compliant with IS 456:2000 and SP 34, especially regarding fundamental design parameters. However, some specific details regarding steel percentages, cross-section area usage, and steel curtailment are missing from the general notes. The grade of concrete compliance cannot be fully verified without environmental exposure details.